

Text-to-Image Generation with Stable Diffusion & FastAPI

This project is a simple web application that generates images from text prompts using the Stable Diffusion v1.5 model. It features a real-time, chat-style interface where users can input a description, and the AI will generate and display the corresponding image.

Features

- **Interactive Web UI:** A clean, chat-like interface to enter prompts and view generated images.
- **High-Quality Image Generation:** Utilizes the runwayml/stable-diffusion-v1-5 model for generating 512x512 images.
- **Performance Metrics:** Displays the time taken for each image generation request.
- **Download Option:** Allows users to download the generated images directly from the UI.
- **FastAPI Backend:** Built with a modern, fast (high-performance) web framework for Python.

Tech Stack

- **Backend:**
 - Python 3.8+
 - FastAPI
 - Diffusers (Hugging Face)
 - PyTorch
 - Jinja2
- **Frontend:**
 - HTML
 - Tailwind CSS
 - JavaScript (Vanilla)

Setup and Installation

Prerequisites

- Python 3.8 or newer
- pip package manager
- For GPU acceleration (recommended):
 - An NVIDIA GPU with CUDA support
 - CUDA Toolkit and cuDNN installed

Steps

1. **Clone the repository:**

```
git clone  
[https://github.com/your-username/your-repo-name.git](https://github.com/your-username/your-repo-name.git)  
cd your-repo-name
```

2. **Create and activate a virtual environment:**

```
# For Windows  
python -m venv venv  
.\venv\Scripts\activate  
  
# For macOS/Linux  
python3 -m venv venv  
source venv/bin/activate
```

3. Install the required Python packages:

Create a requirements.txt file with the following content:

```
fastapi  
uvicorn[standard]  
jinja2  
python-multipart  
torch  
diffusers  
transformers  
accelerate
```

Then, install the packages:

```
pip install -r requirements.txt
```

Note: For GPU support, ensure you install the correct version of PyTorch for your CUDA setup. Visit the [PyTorch website](#) for specific instructions.

4. Organize your project files:

Ensure your directory structure looks like this:

```
.  
├── templates/  
│   └── index.html  
├── main.py  
└── requirements.txt
```

5. **Run the FastAPI server:**

```
uvicorn main:app --reload
```

The --reload flag automatically restarts the server when you make changes to the code.

Usage

1. Once the server is running, open your web browser and navigate to <http://127.0.0.1:8000>.
2. Type a descriptive prompt into the input box at the bottom of the page (e.g., "An astronaut riding a horse on Mars").
3. Click the "Generate" button or press Enter.
4. The AI will process your request, and the generated image will appear in the chat window.
5. You can then download the image using the "Download" button below it.

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